

NIKHIL GOUD KALALI

Software Engineer · AI/Agentic Systems · Backend Engineering
+91 6305443501 | nikhilgoud8837@gmail.com | [linkedin.com/in/nikhil-goud-kalali-465919213](https://www.linkedin.com/in/nikhil-goud-kalali-465919213)

PROFESSIONAL SUMMARY

Results-driven Software Engineer with hands-on experience designing and deploying production-grade Agentic AI systems, multi-agent orchestration platforms, and enterprise-grade backend services. Skilled in Java, Spring Boot, and Spring AI, with deep expertise in LLM integration, RAG architectures, and real-time audio processing. Passionate about building scalable, intelligent automation solutions that deliver measurable business value.

PROFESSIONAL EXPERIENCE

Software Engineer | Vision Waves

Sep 2024 – Present

Indore, India

Project 1: Agentic AI Platform (SaaS)

- Architected a scalable **SaaS Agentic AI platform** supporting **Conversational, Flow-Driven, and Task agent** archetypes for **multi-agent orchestration** and fully autonomous execution.
- Enabled **multi-channel interactions** across **chat, voice, phone, and avatar** touchpoints.
- Designed **SwitchAgent** — an agent orchestration engine for intelligent agent-to-agent routing and modular sub-agent composition, reducing routing latency by ~30%.
- Implemented a full agent governance lifecycle and versioning system (DRAFT → APPROVAL → ACTIVE → RETIRED) with configurable approval, scheduled validity windows, and safe version iteration via DRAFT_VERSION cloning — enabling controlled, auditable rollout from local to production.
- Instrumented **execution trace observability** (per-step timing, token/cost breakdown, provenance log), driving ~20% operational cost reduction.
- Worked within a platform supporting Google A2A v0.3 peer-agent delegation (Agent Card discovery, async push notifications, bearer auth) and MCP (Model Context Protocol) integration with STDIO/SSE transports, connecting 8+ enterprise systems (Salesforce, ServiceNow, GitHub, MySQL, Cassandra) via credential-isolated adapters.
- Built **MCP (Model Context Protocol)** integration with **STDIO and SSE transports**, connecting 8+ enterprise systems (Salesforce, ServiceNow, GitHub, MySQL, Cassandra) via credential-isolated adapters.
- Contributed to long-term episodic memory (conversation summarization), reused RAG infrastructure for contextual retrieval, and supported Conversational Handoff for context-preserving agent-to-agent escalation.
- Contributed to a Skills Library enabling portable SKILL.md / .zip reasoning modules reusable across agents.

Backend Engineering & Infrastructure

- Built production REST APIs using **Java, Spring Boot, and Spring AI** for high-throughput AI-driven services.
- Implemented **Speech-to-Text (STT)** and **Text-to-Speech (TTS)** pipelines on **Google Cloud Platform (GCP)** for real-time voice interactions.
- Developed **real-time audio processing** modules and **state-machine workflow handlers** for stateful multi-turn conversational flows.

TECHNICAL SKILLS

Languages: Java

Frameworks & Libraries: Spring Boot, Spring AI

AI / Agentic Systems: MCP (Model Context Protocol), Google A2A v0.3, RAG, LLM Integration, Multi-Agent Orchestration, OPA Policy Engine, Vector Databases

Databases: SQL, PostgreSQL

Developer Tools: Claude AI, Cursor, Postman, Git

Concepts: State Machine Design, Real-Time Audio Processing, Semantic Pre-Filtering, Agent Governance

EDUCATION

B.Tech — Computer Science and Engineering

2020 – 2024

Malla Reddy College of Engineering and Technology CGPA: 7.78 / 10

Intermediate (MPC)

2018 – 2020

Sandeepani Junior College CGPA: 8.24 / 10